

Machine Operation Manual — AI Factory Fleet

This manual defines normal operating ranges and out-of-range thresholds for every machine type in the fleet (CNC, Press, Injection Molder). Warning thresholds indicate a reading that requires operator attention; danger thresholds indicate a reading that requires immediate action (see the Preventive Maintenance SOP and Safety & Emergency Guidelines documents for required responses).

Section 1: CNC

Rated power draw: 2860 W. Readings below are per-15-minute sensor sample.

Sensor	Normal (mean)	Warning threshold	Danger threshold
Temperature	301.0 K	≥ 305.0 K	≥ 308.0 K
Vibration	1.8 mm/s	≥ 2.6 mm/s	≥ 3.2 mm/s
Pressure	5.5 bar	≥ 6.5 bar	≥ 7.2 bar
Torque	40.0 Nm	≥ 60.0 Nm	≥ 75.0 Nm

Vibration guidance: normal running vibration for a CNC is approximately 1.8 mm/s. A sustained reading at or above 2.6 mm/s is a WARNING condition and should be logged and monitored. A sustained reading at or above 3.2 mm/s is a DANGER condition — vibration this high is dangerously elevated and indicates likely bearing wear, misalignment, or an imminent mechanical failure.

Temperature guidance: normal operating temperature is approximately 301.0 K. 305.0 K or above is a WARNING condition; 308.0 K or above is a DANGER condition indicating possible overheating or lubrication failure.

Pressure guidance: normal operating pressure is approximately 5.5 bar. 6.5 bar or above is a WARNING condition; 7.2 bar or above is a DANGER condition indicating a possible seal or valve fault.

Section 2: Press

Rated power draw: 4200 W. Readings below are per-15-minute sensor sample.

Sensor	Normal (mean)	Warning threshold	Danger threshold
Temperature	304.0 K	≥ 309.0 K	≥ 312.8 K
Vibration	2.6 mm/s	≥ 3.8 mm/s	≥ 4.7 mm/s
Pressure	12.0 bar	≥ 14.4 bar	≥ 16.2 bar
Torque	55.0 Nm	≥ 79.0 Nm	≥ 97.0 Nm

Vibration guidance: normal running vibration for a Press is approximately 2.6 mm/s. A sustained reading at or above 3.8 mm/s is a WARNING condition and should be logged and monitored. A sustained reading at or above 4.7 mm/s is a DANGER condition — vibration this high is dangerously elevated and indicates likely bearing wear, misalignment, or an imminent mechanical failure.

Temperature guidance: normal operating temperature is approximately 304.0 K. 309.0 K or above is a WARNING condition; 312.8 K or above is a DANGER condition indicating possible overheating or lubrication failure.

Pressure guidance: normal operating pressure is approximately 12.0 bar. 14.4 bar or above is a WARNING condition; 16.2 bar or above is a DANGER condition indicating a possible seal or valve fault.

Section 3: Injection Molder

Rated power draw: 3500 W. Readings below are per-15-minute sensor sample.

Sensor	Normal (mean)	Warning threshold	Danger threshold
Temperature	308.0 K	≥ 314.0 K	≥ 318.5 K
Vibration	1.4 mm/s	≥ 2.0 mm/s	≥ 2.5 mm/s
Pressure	18.0 bar	≥ 21.6 bar	≥ 24.3 bar
Torque	30.0 Nm	≥ 46.0 Nm	≥ 58.0 Nm

Vibration guidance: normal running vibration for a Injection Molder is approximately 1.4 mm/s. A sustained reading at or above 2.0 mm/s is a WARNING condition and should be logged and monitored. A sustained reading at or above 2.5 mm/s is a DANGER condition — vibration this high is dangerously elevated and indicates likely bearing wear, misalignment, or an imminent mechanical failure.

Temperature guidance: normal operating temperature is approximately 308.0 K. 314.0 K or above is a WARNING condition; 318.5 K or above is a DANGER condition indicating possible overheating or lubrication failure.

Pressure guidance: normal operating pressure is approximately 18.0 bar. 21.6 bar or above is a WARNING condition; 24.3 bar or above is a DANGER condition indicating a possible seal or valve fault.